An Interactive Dual-Mode Catamaran Prototype: Project Nereus V3.1

Content

1. The Brief & Problem Statement

1.1 Context and Problem Definition

The primary objective of this project is to conceptualize, design, and develop a small-scale, remote-controlled marine vessel capable of assisting in local water quality monitoring, environmental surveying, or aquatic payload transport. Many existing hobbyist or entry-level commercial remote-controlled (RC) boats utilize a standard monohull design, which inherently lacks the lateral stability required to carry delicate sensors or steady payloads. Furthermore, traditional RC boats are single-purpose (solely focused on propulsion), making them inadequate for interactive tasks such as deploying and retrieving underwater equipment. Therefore, there is a tangible engineering need for a highly stable, multi-functional prototype that can navigate smoothly across water bodies while providing a reliable mechanical system for interacting with the aquatic environment.

1.2 Proposed Solution and Inspiration: "Project Nereus"

"Project Nereus" is an advanced, fully functional catamaran prototype engineered to demonstrate how a dual-hull structure can be seamlessly integrated with interactive mechanical systems. The core operational philosophy revolves around a "Dual-Mode System" managed by a custom-written operating system (Nereus OS), allowing the user to seamlessly switch between sailing and crane operations using a single dual-axis joystick.

The conceptualization of the central payload deployment system was heavily inspired by the video game *Dave the Diver*, specifically the mechanics of the underwater elevator escape pod. Drawing from this, we envisioned practical, real-world applications for a submersible elevator cage. For instance, in marine harvesting, fishermen could place their caught nets directly into the submerged elevator. This ensures the fish remain in their natural ocean environment, significantly extending their survival time compared to being stored on a dry deck, while the surrounding iron cage protects them from predators. Alternatively, scaled-up versions of this system could serve as temporary protective shark-cages for divers working in dangerous waters. While our current prototype cage is open at the top, it serves as a proof of concept that can easily be enclosed or modified for actual industrial applications. As illustrated in Figure 1, this early conceptualization depicts the twin hulls and the central A-frame gantry lowering the *Dave the Diver*-inspired submersible cage.

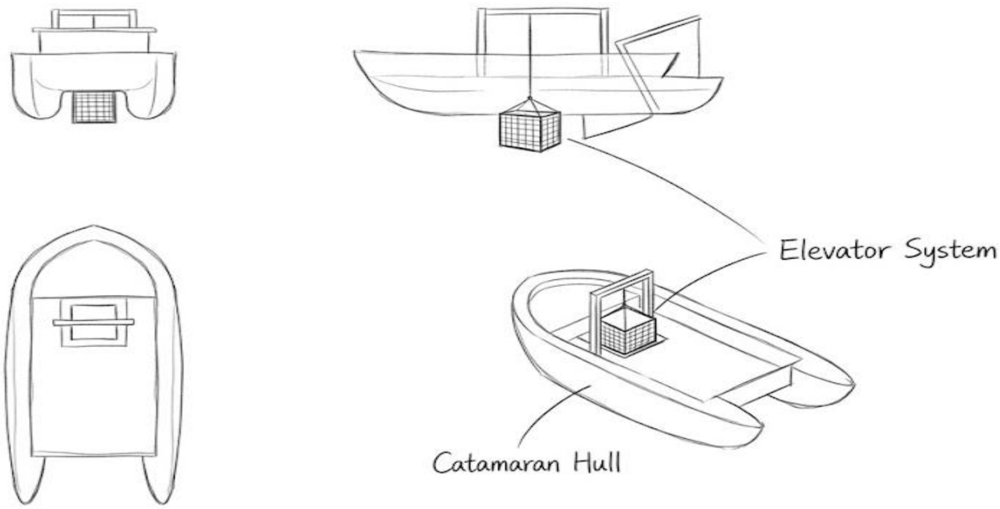

Figure 1: Early Minimalist Concept Sketch of the Project

2. Project Development Workflow & Iteration

The development of Project Nereus adhered to a rigorous, iterative prototyping pipeline, focusing heavily on overcoming technical bottlenecks, balancing mechanical robustness, and ensuring electronic safety.

The workflow commenced with conceptual 3D modeling in SolidWorks to define the vessel's structural anatomy. Before any physical assembly, a critical Tinkercad simulation phase was conducted to validate the circuit logic and prevent hardware damage. Subsequently, hardware selection underwent a significant evolution. We upgraded from standard 9V batteries to a robust 7.4V Lithium-Polymer (Li-Po) battery, integrated an industrial-grade ZK-5AD motor driver, and refined our servo selection based on torque requirements. Following digital validation, the optimized parts were 3D printed. The final phase involved resolving complex software bugs and coding a non-blocking architecture to handle asynchronous screen rendering. This comprehensive evolutionary journey, from initial CAD drafting to final physical integration, is visually summarized in Figure 2, which highlights the iterative problem-solving approach taken throughout the project.

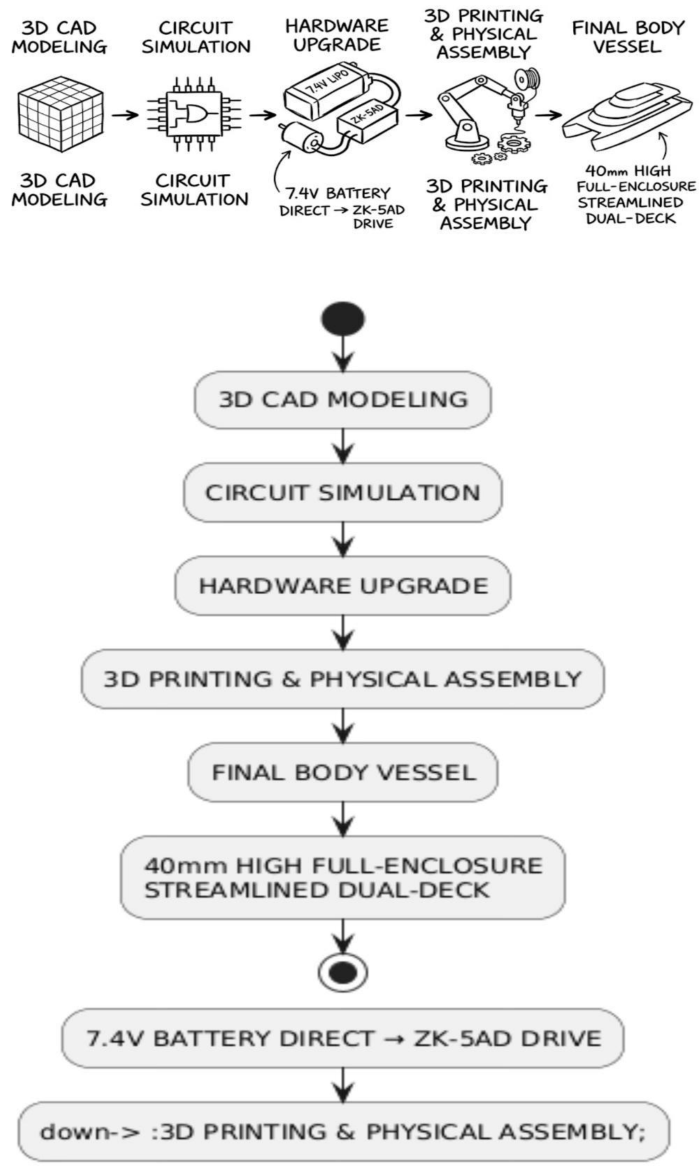

Figure 2: Flowchart of Project Development and Iteration

2.1 Project Schedule

The development of Project Nereus was executed over a four-week period. The following table 1 breaks down the key milestones and tasks completed during each phase:

Table 1: Course Project Schedule for Prototyping

Phase	Duration	Key Tasks and Milestones
Week 1	**Concept & Initial Design**	Defined the project scope inspired by the escape pod mechanics in *Dave the Diver*. Developed the first iteration of the vessel, initially conceptualized as a compact skiff rather than a full-scale ship.
Week 2	**Functional Planning & Simulation**	Finalized the dual-mode feature set. Conducted extensive circuit simulations in Tinkercad to validate the logic of mode-switching and I2C communication before physical assembly.
Week 3	**Hardware Assembly & Troubleshooting**	Focused on physical circuit construction. Significant time was invested in resolving persistent power configuration issues (USB crashes and common ground faults). Successfully achieved full hardware-software synergy.
Week 4	**Refinement, 3D Printing & Testing**	Optimized the SolidWorks model into the final dual-deck streamlined hull. Printed 7 components and conducted final assembly. Executed field testing to ensure all functional requirements were met.

3. Mechanical Design and Aesthetic Engineering in SolidWorks

As the project evolved from a rudimentary skeleton to a professional marine research vessel, the mechanical design underwent a complete overhaul in SolidWorks to achieve both functional superiority and a refined "Tech-Style" aesthetic.

3.1 Hydrodynamic Pontoons and the Armored Deck

The foundation of the vessel utilizes two streamlined pontoons designed to minimize water resistance. In earlier design iterations, we attempted to protect the electronic components by engineering a 15mm high continuous coaming around a flat main deck. However, to achieve a superior industrial design and better space management, we completely overhauled the structure in the final version. We eliminated the basic coamings and introduced an innovative "dual-deck" architecture: a complete 40mm-high outer hull shell with a streamlined profile was constructed above the base plate, capped by a newly added upper deck. This design creates a spacious, enclosed internal cabin between the two decks, perfectly concealing all messy hardware and wiring.Figure 3 demonstrates the precise fitment of these components that the front of this 40mm-high hull seamlessly integrates a slanted, streamlined bow splash guard. This major structural upgrade not only mitigates the "bow slamming" effect and significantly improves hydrodynamic performance, but also gives the vessel a continuous, sleek exterior that closely aligns with the classic aesthetic of a professional marine vessel.

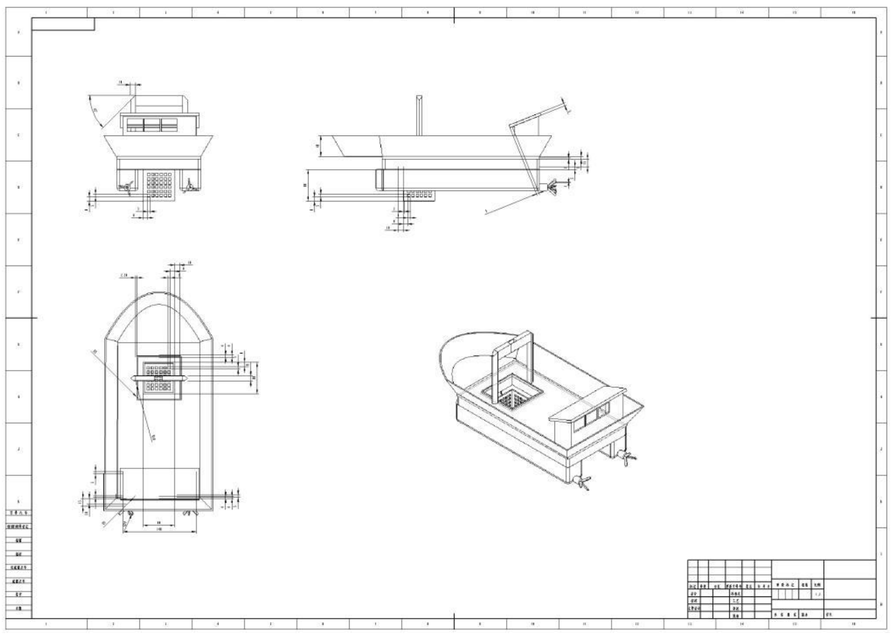

Figure 3: Screenshot of the Final SolidWorks Assembly

3.2 The Retro Cabin and A-Frame Gantry

To house the complex electronics, a hollow, single-slope retro cabin was designed. The cabin features a 3mm thin-wall construction and incorporates an overhanging hardtop canopy with observation windows to enhance the industrial aesthetic. Positioned directly above the moon pool is the A-Frame crane. This structure was designed with heavily chamfered edges to distribute mechanical stress. As captured in Figure 3 above, the SolidWorks assembly demonstrates the precise fitment of these components, particularly highlighting the distance mates applied to the submersible cage to simulate its vertical trajectory without intersecting the hulls.

3.3 Modular Component Design for 3D Printing

To facilitate rapid prototyping and ensure high-precision manufacturing, the mechanical structure of Project Nereus was engineered with a highly modular approach. Rather than attempting to print the vessel as a single complex monolithic block, the design was broken down into seven distinct, interchangeable CAD components.

As illustrated in Figure 4, these individual custom-designed parts include: the flat upper deck plate featuring the moon pool cutout, the 40mm streamlined outer hull shell with the integrated bow splash guard, the lower hydrodynamic pontoon base, the retro-style control cabin, the A-frame gantry structure, the submersible payload cage, and the custom 3-blade marine propeller.

This modularity provided two significant engineering advantages. First, it allowed for rapid individual component iteration during the design phase without the cost and time penalty of re-printing the entire boat. Second, it optimized the 3D printing process by allowing each part to be oriented flat on the print bed, drastically minimizing the need for wasteful breakaway support structures and ensuring a smoother surface finish for the final assembly.

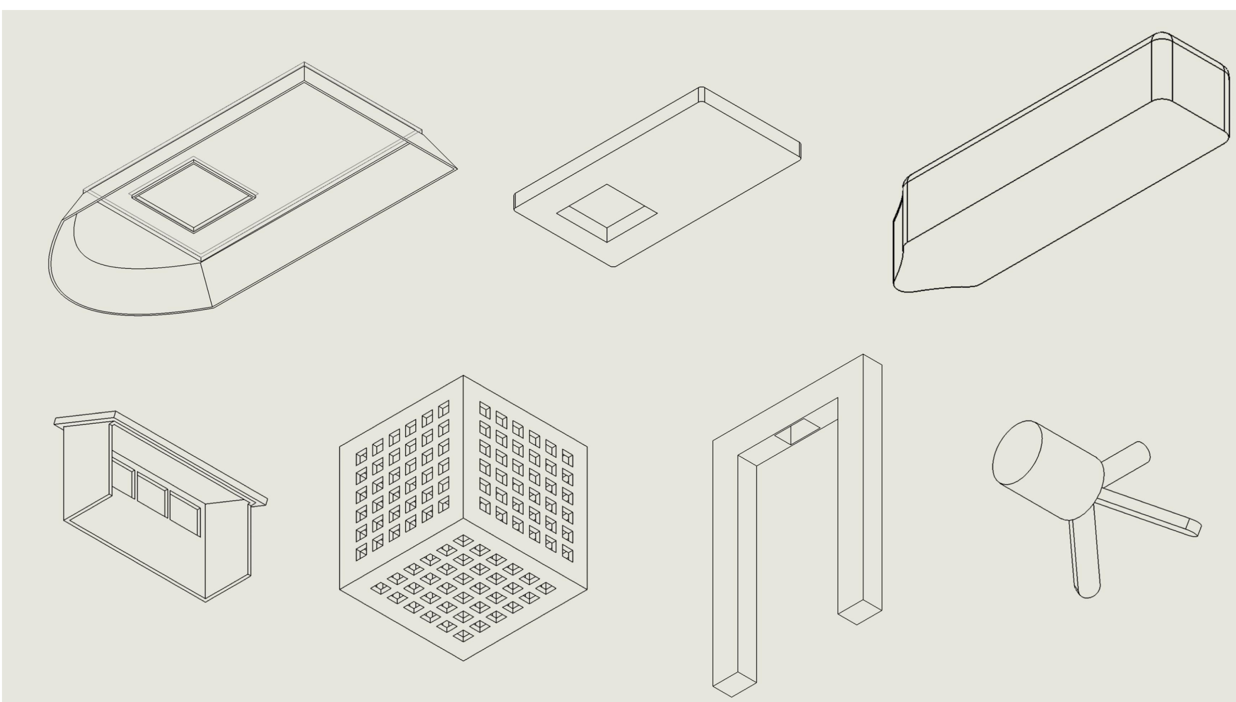

Figure 4: Isometric View of the 7 Custom 3D-Printed Modular Components

4. Hardware Selection and Electronic Architecture

Transitioning from a basic prototype to a high-performance vessel required addressing specific mechanical limitations through careful hardware selection. The current system is a carefully balanced network of high-voltage power components and low-voltage logic controllers.

4.1 Propulsion and the ZK-5AD Integration

The original 9V zinc-carbon battery was replaced with a high-discharge 7.4V Li-Po battery to power the main engine: the JGA25-370 geared DC motor. This motor provides exceptional torque at low RPMs. To control this motor, the ZK-5AD motor driver was selected. During the assembly, we faced a critical decision regarding the power input for the ZK-5AD. Initially considering the VT (Voltage Terminal) pin, we ultimately bypassed it and the adjacent GND pin entirely. Instead, we wired the 7.4V battery directly to the primary red power terminal block of the driver. This direct connection provided a much more stable and robust power delivery for the high-current demands of the JGA25-370 motor.

4.2 Differentiated Servo Configuration

A major hardware optimization involved differentiating the servo motors based on their specific mechanical tasks. Initially, we considered using two identical micro-servos. However, practical testing revealed that the steering mechanism required higher torque and specific angular constraints. Consequently, we integrated a heavy-duty MG996R servo exclusively for the steering rudder. The MG996R's built-in angle limitations and superior torque make it ideal for acting as a stable steering wheel. Conversely, the smaller SG90 micro-servo was retained for the A-Frame winch, as lowering and raising the lightweight cage required precision rather than raw power. The integration of this dual-servo system, alongside the OLED display and ultrasonic sensor, is explicitly mapped out in Figure 5 which details the power distribution and signal pathways.

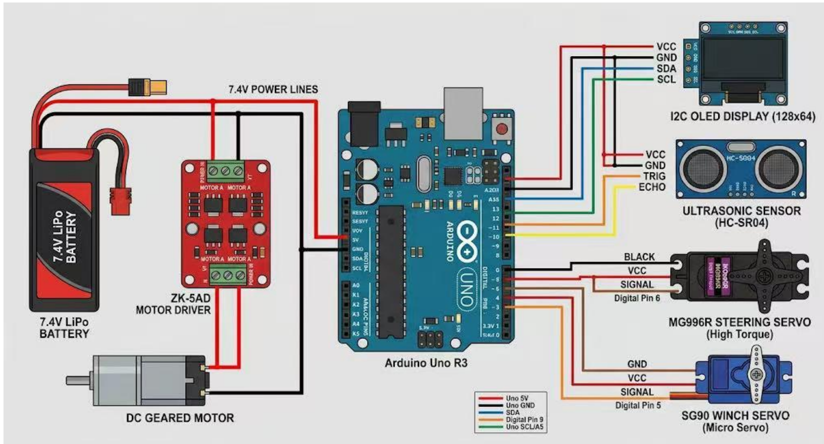

Figure 5: Topology Diagram of the Complete Circuit System

5. Overcoming Experimental Challenges: Troubleshooting and Power Strategy

The physical assembly and testing phase presented significant engineering challenges that required systematic troubleshooting. Highlighting these obstacles is crucial, as resolving them fundamentally shaped the final prototype.

5.1 The USB Overcurrent Crisis

The most critical hardware failure occurred when attempting to run the motors and servos simultaneously while connected to the computer. The system crashed, triggering an "Unrecognized USB Device" error on the host PC. We diagnosed this as a severe overcurrent protection trigger; the instantaneous current drawn by the JGA25-370 motor and the MG996R servo drastically exceeded the Arduino's 5V regulator limit and the USB interface's 500mA capacity, causing the computer's motherboard to physically sever the connection to protect itself.

5.2 Implementing "Water and Electricity Separation"

To resolve this, we engineered a strict "Power Isolation" strategy on the breadboard. The system was divided into a high-current zone and a low-current logic zone. As previously mentioned, the

7.4V battery was routed directly to the red power terminal of the ZK-5AD to handle the main propulsion. We then established a separate 5V rail to power the SG90 winch servo, the ultrasonic sensor, and the OLED display. The most vital step in this resolution was establishing a "Common Ground" by explicitly linking the ground of the 7.4V battery, the ZK-5AD driver, and the Arduino. Without this shared reference voltage, the PWM signals would remain unrecognizable. Figure 6 provides a close-up view of this implemented solution on the breadboard, clearly showing the power isolation and the crucial common ground connections that saved the microcontroller from frying.

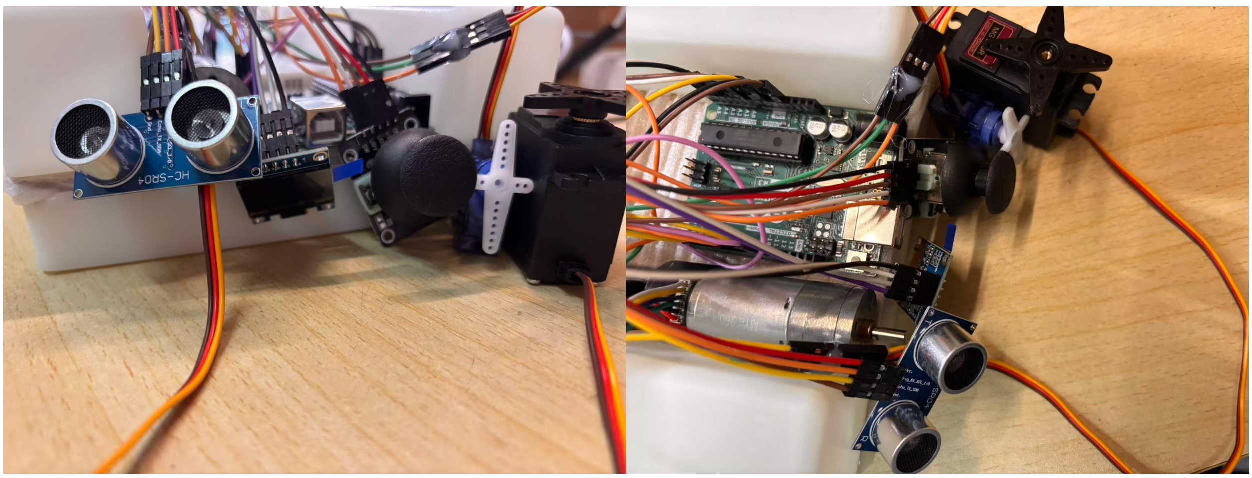

Figure 6: Physical Wiring on the Breadboard

6. Software Design and Tinkercad Simulation

The software architecture and pre-assembly simulations were pivotal in ensuring the prototype functioned smoothly without physically destroying components during trial and error.

6.1 Tinkercad Circuit Simulation and Logic Validation

Before wiring the physical 7.4V battery, we simulated the entire Nereus OS logic in Tinkercad. Since the platform lacks the specific ZK-5AD driver and SSD1306 OLED, we utilized an L293D H-bridge and a 16x2 I2C LCD as equivalent components to test our PWM and I2C telemetry logic. During initial simulations, we encountered virtual "component explosion" errors because the simulated motor pulled current directly from the microcontroller. By correctly routing external power through the L293D—simulating our "Power Isolation" strategy—we eliminated these errors. As showcased in figure 7, the simulation successfully validated the dual-mode logic. The screenshots demonstrate the system's successful boot sequence and the seamless transition between "SAIL" mode (main motor engaged at 100% power, telemetry active) and "CRANE" mode (main motor safely locked at 0% power, winch servo active), proving the software architecture was completely sound before any physical wiring was attempted.

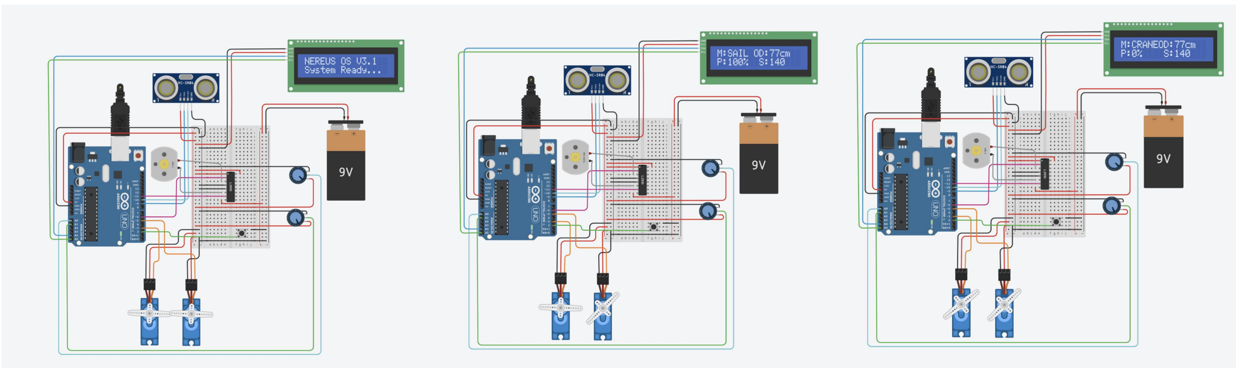

Figure 7: Screenshot of Tinkercad Simulation Running

6.2 Nereus OS V3.1 and the Screen Blocking Bug

The brain of the vessel is powered by a custom Arduino C++ program utilizing a robust State Machine. The analog joystick serves a dual purpose: providing X/Y values and acting as a digital pushbutton to toggle between "SAIL" and "CRANE" modes. We programmed a critical safety mechanism: when the mode switches, the code executes analogWrite(DRIVER_D0_PIN, 0); analogWrite(DRIVER_D1_PIN, 0);, instantly cutting the main engine. This prevents the boat from accelerating uncontrollably while the operator focuses on lowering the cage.

A major software challenge emerged with the integration of the I2C OLED display. In an earlier iteration, if the screen wires became loose due to vibrations, the display.begin() function failed, triggering a for(;;) infinite loop that completely froze the Arduino, rendering the boat dead in the water. We rewrote the architecture in V3.1 to include a "Blind-Drive" fail-safe. If the screen fails to initialize, the code sets a boolean flag (hasScreen = false), bypasses the rendering functions, and forces the motor control loops to continue running. Furthermore, to prevent the display.display() function from stuttering the servo movements, we implemented a non-blocking asynchronous timer using the millis() function, restricting screen updates to every 150 milliseconds. The intricate flow of this state machine and the fail-safe branches are logically mapped out in Figure 8.

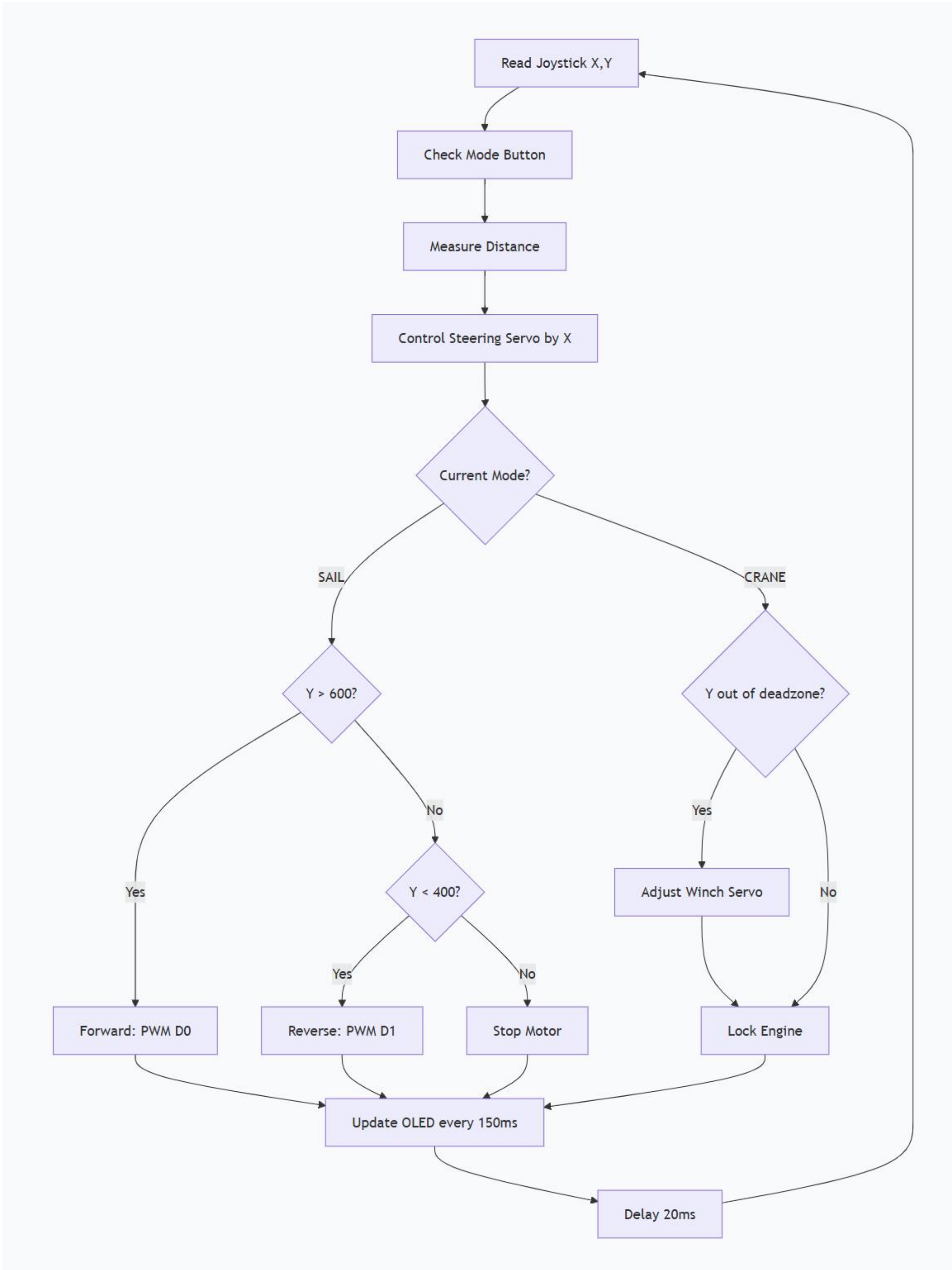

Figure 8: Program Logic Flowchart of Nereus OS v3.1

7. Interactive Aesthetics and Industrial Design

The final deliverable bridges the gap between mechanical engineering and industrial design. The structural elements are not merely functional; they represent a cohesive product identity. The integration of the OLED display transformed the remote control experience, providing a continuous stream of telemetry data including power output (PWR: %), steering angles, and radar proximity from the HC-SR04.

The physical vessel, featuring the aggressive bow splash guard, the angled cabin hiding the complex wiring, and the exposed A-frame crane, embodies a rugged, scientific aesthetic. It looks less like a hobbyist toy and more like a scaled-down industrial marine tool. The transformation from a bare-bones electronics testbed into a visually striking, mission-ready catamaran is showcased in Figure 9, displaying the vessel's polished exterior and operational stance.

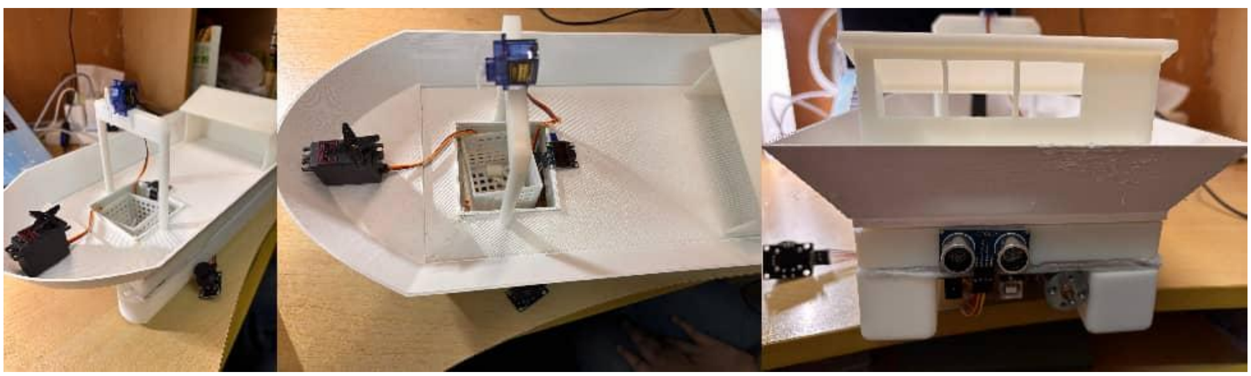

Figure 9: Real-world Photo

8. Bill of Materials (BOM)

The construction of the Project Nereus V3.1 prototype relied on a carefully curated set of electronic and mechanical components, each selected to address specific engineering requirements identified during our four-week iterative development process. Our hardware selection strategy prioritized a deliberate balance between performance, reliability, and resource efficiency, with targeted upgrades implemented to resolve initial power delivery and torque limitations. The finalized component list（Table 2）, presented in full on the following page, encapsulates all hardware iterations and optimizations made throughout the project, ensuring seamless operation of both the dual-hull propulsion system and the precision payload deployment mechanism.

Table 2: Bill of Materials for Project Prototype

Designation	Quantity	Component Description & Specification	Total Cost
U1	1	Arduino Uno R3 (Main Logic Controller)	0
BAT1	1	7.4V 2S Li-Po Battery (High-Discharge Power Source)	0
U2	1	ZK-5AD High-Power Motor Driver Module	0
M1	1	JGA25-370 Geared DC Motor (Main Engine)	0
SERVO_ST	1	MG996R High-Torque Servo (Steering Rudder Actuator)	0
SERVO_WI	1	SG90 Micro Servo (A-Frame Winch Actuator)	0
DISP1	1	0.96-inch I2C SSD1306 OLED Display	0
DIST1	1	HC-SR04 Ultrasonic Distance Sensor	0
JOY1	1	Dual-Axis Analog Joystick with Pushbutton	0
PROP	1	Custom 3D-Printed 3-Blade Marine Propeller	0
3D Parts	7	Custom-designed Hull, Gantry, and Fittings	314

9. Conclusion

Project Nereus V3.1 represents a comprehensive engineering journey, successfully transitioning from a conceptual idea inspired by *Dave the Diver* into a highly capable, dual-mode marine prototype. The project's success is rooted in the rigorous troubleshooting of severe hardware and software issues. By analyzing and overcoming the USB overcurrent crashes through power isolation, and by reprogramming the software to bypass blocking loops using asynchronous millis() logic, we ensured the vessel's reliability. Furthermore, distinguishing our actuators—using the heavy-duty MG996R for steering and the precise SG90 for the payload elevator—maximized mechanical efficiency. The final vessel stands as a testament to iterative design, proving to be a stable, reliable, and aesthetically polished product ready to demonstrate interactive aquatic payload deployment in real-world environments.

Reflecting on the entire development process, we have learned that rigorous hardware power management is the absolute priority when integrating high-torque actuators with sensitive logic controllers. Although we successfully overcame significant setbacks, several key improvements would be implemented in future iterations to transition this prototype into a commercial-grade tool.

First, we would prioritize the use of high-quality waterproof gaskets and marine-grade sealants to fully hermeticize the dual-deck hull compartment. This would enable the vessel to move beyond its current "non-waterproof" demonstration phase and become a functional entity capable of long-term operations in natural, unpredictable water environments. Secondly, while the Dave the Diver-inspired open elevator cage performed reliably during testing, a future version would incorporate an automated top-hatch mechanism. This addition would ensure that payloads or aquatic samples are fully secured and protected from predators or environmental turbulence during transport.

Ultimately, this project has not only significantly enhanced our hands-on engineering and troubleshooting skills but also deepened our understanding of the delicate balance between sophisticated industrial aesthetic design and robust, hardcore engineering logic.